

$$\Rightarrow \phi_C = -4.0 \text{ J/kg}$$

11 B

$$C_{rms} = \sqrt{\frac{3RT}{M}} \propto \sqrt{T} \propto \sqrt{PV} \quad \text{since } PV = nRT$$

$$\frac{C_{rms2}}{C_{rms1}} = \sqrt{\frac{P_1 V_1}{P_2 V_2}} = \sqrt{\frac{2(2)}{1(1)}} = 2$$

12 D

If the gases are ideal $PV = nRT = nR(\theta + 273)$